

ISyE 6669
2020 Spring Final Exam Practice

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Problem 1. Which of the following sets is convex for any positive integer n ?

A $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i \geq -1\}.$

B $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i = -1\}.$

C $X = \{x \in \mathbb{R}^n : \max_{i=1,\dots,n} x_i \leq -1\}.$

Solution: C

Consider $n = 2$. A schematic of $\max_{i=1,2} x_i$ will be a vertical ray starting at -1 and going to $-\infty$, meeting with a horizontal ray starting at -1 and going to $-\infty$. A convex set would have a line drawn between any two points within the set still contained. In order for this to be true, the points contained within the set should be below the horizontal ray and left of the vertical ray. This corresponds to option C.

Problem 2. Which of the following functions is convex on the specified domain?

A $f(x) = 5 - |x|$ on $x \leq 0$

B $f(x) = x - |x|$ on $x \in [-7, 1]$

C $f(x) = -|x|$ on $x \in \mathbb{R}$

Solution: A

For A, $f(x)$ amounts to $5 + x$ for the entirety of the specified domain ($x \leq 0$). Being a linear function, it is convex and is thus the correct response. B is concave because it is $2x$ when $x \leq 0$ and 0 when $x > 0$. C is a reflection about the horizontal axis of the convex absolute value function. Thus, it is concave.

Problem 3. Which of the following statements is true about the extreme points and extreme rays?

A Any polyhedron has an extreme ray.

B If a nonempty polyhedron does not have any extreme point, then it is unbounded.

C For a nonempty polyhedron, any point in the polyhedron can be written as the non-zero sum of at least two different extreme points and some extreme rays.

Solution: B

A bounded polyhedron does not have an extreme ray. Thus, A and C are incorrect. The statement B is correct.

Problem 4. Consider a constrained optimization problem

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & g(x) \geq 0, \\ & h(x) = 0, \\ & x \in X. \end{aligned}$$

Which of the following problems is a relaxation of this problem?

A $\min\{f(x) + h(x) : g(x) \geq -1, x \in X\}$

B $\max\{f(x) + g(x) : h(x) = 10, x \in X\}$

C $\min\{-g(x) + h(x) : x \in X\}$

Solution: A

For A, the objective is considered a relaxation of the objective in the problem statement because they are equal to each other (as $h(x) = 0$). The constraint $g(x) \geq 0$ from the problem statement is a subset of $g(x) \geq -1$. Thus, A is a relaxation of the given problem. For B, because $g(x) \geq 0$, the objective is not a relaxation of objective in the problem statement. For C, it is not possible to compare the objective $f(x)$ of the problem statement with the objective of C, $-g(x) + h(x)$. Thus, we cannot say that the objective of C is a relaxation of the objective in the problem statement with the given information.

Problem 5. Consider the following linear program for $(x_1, x_2, x_3) \in \mathbb{R}^3$:

$$\begin{aligned} \min \quad & 3x_1 + 5x_2 + 7x_3 \\ \text{s.t.} \quad & -x_1 + 3x_2 = 5, \\ & 2x_1 - x_2 + 3x_3 \geq 6, \\ & x_3 \leq 7, \\ & x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.} \end{aligned}$$

The dual of the above linear program can be written as:

A

$$\begin{array}{ll}
\max & 5y_1 + 6y_2 + 7y_3 \\
\text{s.t.} & -y_1 + 2y_2 \leq 3, \\
& 3y_1 - y_2 \geq 5, \\
& 3y_2 + y_3 = 7, \\
& y_1 \text{ free}, y_2 \leq 0, y_3 \geq 0.
\end{array}$$

B

$$\begin{array}{ll}
\max & 5y_1 + 6y_2 + 7y_3 \\
\text{s.t.} & -y_1 + 2y_2 \geq 3, \\
& 3y_1 - y_2 \leq 5, \\
& 3y_2 + y_3 = 7, \\
& y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0.
\end{array}$$

C

$$\begin{array}{ll}
\max & 5y_1 + 6y_2 + 7y_3 \\
\text{s.t.} & -y_1 + 2y_2 \leq 3, \\
& 3y_1 - y_2 \geq 5, \\
& 3y_2 + y_3 = 7, \\
& y_1 \text{ free}, y_2 \geq 0, y_3 \leq 0.
\end{array}$$

Solution: C

Recall that for a given primal problem $\left\{ \min c^T x : Ax = b, x \geq 0 \right\}$, its dual is $\left\{ \max b^T y : y^T A \leq c^T, y \text{ free} \right\}$, with further rules determining the sign of the dual variables based on the equalities or inequalities of the primal problem. Here, we have that $c = [3, 5, 7]^T$, $b = [5, 6, 7]^T$, and $A = \begin{bmatrix} -1 & 3 & 0 \\ 2 & -1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$. The first primal constraint is an equality. Thus, its corresponding dual variable, y_1 , is free. The second primal constraint involves a $\geq$ inequality. Thus, its corresponding dual variable, y_2 , is ≥ 0 . The third primal constraint involves a $\leq$ inequality. Thus, its corresponding dual variable, y_3 , is ≤ 0 . The first primal variable, x_1 , is ≥ 0 , so its corresponding dual constraint will involve a $\leq$ inequality. The second primal variable, x_2 , is ≤ 0 , so its corresponding dual constraint will involve a $\geq$ inequality. The third primal variable, x_3 , is equal to 0, so its corresponding dual constraint will be an equality. Putting all this together, we arrive at C as the solution.

Problem 6. Consider the following linear program:

$$(P) \quad \min \quad c^T x$$

$$\text{s.t.} \quad Ax \leq 0,$$

$$0 \leq x \leq d$$

Select all correct statements.

- A The problem (P) is possible to be infeasible.
- B If the problem (P) is unbounded, then its dual is unbounded.
- C If the problem (P) has a finite optimal, then its dual has a finite optimal.

Solution: The correct answer is C.

For option A, since $x \geq 0$ this will guarantee that problem P having at least 1 feasible solution. Option B, if the primal is unbounded, then the dual is infeasible. Hence Option C is obviously the correct answer.

Problem 7. Consider the robust linear constraint for $x \in \mathbb{R}^n$:

$$a_1^\top x + a_2^\top y \geq b, \quad \forall a \in \mathcal{A} = \{a_1, a_2 \in [0, 1]^n : C_1 a_1 \leq d_1, C_2 a_2 \leq d_2\}$$

Which of the following is the correct reformulation.

- A $C_1 a_1 + C_2 a_2 - d_1 - d_2 \geq b$.
- B $d_1^\top z + d_2^\top w + e^\top u + e^\top v + b \leq 0, \quad x + C_1^\top z + u \geq 0, \quad y + C_2^\top w + v \geq 0, \quad z, w, u, v \geq 0$, where $e = (1, \dots, 1)^\top \in \mathbb{R}^n$.
- C $\sum_{i=1}^n x_i + y_i \geq b$.

Solution: B

It is obvious that option A is not the answer as the set has a $\leq$ in $C_1 a_1 \leq d_1$ but the original robust constraint has $\geq b$. It is also obvious that option C is not correct since it is not covering the coefficient a_1 and a_2 limits even on the index $i = 1, \dots, n$. Thus the remaining option is B.

In order to understand why B is the correct answer, you may go through the following exercise:

From Module 18, Lesson 3, we can rewrite the robust linear constraint as

$$\min_{a_1, a_2} \quad a_1^\top x + a_2^\top y \geq b$$

$$\text{s.t.} \quad C_1 a_1 \leq d_1$$

$$C_2 a_2 \leq d_2$$

$$a_1 \in [0, 1]^n$$

$$a_2 \in [0, 1]^n.$$

We note that we can rewrite $a \in [0, 1]^n$ as $a \leq e$ and $a \geq 0$, where $e = (1, \dots, 1)^T$. Thus, we have

$$\begin{aligned} \min_{a_1, a_2} \quad & a_1^T x + a_2^T y \geq b \\ \text{s.t.} \quad & C_1 a_1 \leq d_1 \\ & C_2 a_2 \leq d_2 \\ & a_1 \leq e \\ & a_2 \leq e \\ & a_1, a_2 \geq 0, \end{aligned}$$

where $e = (1, \dots, 1)^T$. Now, recall that for a given primal problem $\left\{ \min c^T x : Ax = b, x \geq 0 \right\}$, its dual is $\left\{ \max b^T y : y^T A \leq c^T, y \text{ free} \right\}$, with further rules determining the sign of the dual variables based on the equalities or inequalities of the primal problem.

For this problem, we have that $A = \begin{bmatrix} C_1 & 0 \\ 0 & C_2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. We allow the dual variables to be $y = (z, w, u, v)^T$. We also note that because all of our primal constraints are $\leq$, all of our dual variables are also $\leq$. We can then arrive at a dual solution of

$$\begin{aligned} \max_{a_1, a_2} \quad & d_1^T z + d_2^T w + e^T u + e^T v \geq b \\ \text{s.t.} \quad & C_1^T z + u \leq x \\ & C_2^T w + v \leq y \\ & z, w, u, v \leq 0. \end{aligned}$$

Because we prefer to have our variables $\geq$, we can just multiply by them by -1 wherever they appear. We can then arrive at our final solution by ignoring the min in our constraint as applicable and rearranging terms to arrive at

$$\begin{aligned} d_1^T z + d_2^T w + e^T u + e^T v + b &\leq 0 \\ x + C_1^T z + u &\geq 0 \\ y + C_2^T w + v &\geq 0 \\ z, w, u, v &\geq 0. \end{aligned}$$

This corresponds to B.

Problem 8. Consider a simple cutting stock problem, where a big roll has width 7 and demand

takes widths 3 and 2. How many different columns are there in the cutting stock problem formulation?

- A 6
- B 7
- C 10

Solution: The correct answer is B.

The answer to this question relies on the problem formulation. One can use a column vector A_j to represent a feasible cutting pattern p_j . The i -th component of $A_j(a_{ij})$ corresponds to the number of pieces of length s_i cut in one roll of configuration p_j . For p_j to be a feasible cutting pattern, the elements of A_j must all be non-negative integers and the sum of them must be at most equal to the Width. This can be represented as $(\sum_{i=1}^m a_{ij} \leq \text{Width})$. That being said, since Width is 7 then there are 7 columns in the formulation.

Problem 9. Which of the following statements are true about solving a linear program by the column generation algorithm?

- A The column generation algorithm will continue if all the columns are added
- B To start solving the problem through the column generation algorithm, we need to know all the possible patterns
- C The optimal solution of Restricted Master Problem in the column generation algorithm is always a basic feasible solution of Master Problem

Solution: The correct answer is C.

For option A, one of the Column Generation stopping criteria is to stop when all columns are added. For option B, if one would know all the combinations then there would be no need to use any Mathematical Programs to find the optimal one as a simple arithmetic would be efficient enough.

Problem 10. For the polytope $P = [-1, 0]^n$, the extreme point representation of any point $x \in P$ can be written as $x = \sum_{i=1}^n -\lambda_i e_i$, $\sum_{i=1}^n \lambda_i = 1$, $\lambda_i \geq 0$, $i = 1, \dots, n$ where e_i is a vector with the i -th component being 1 and all other components being 0.

- A True
- B False

Solution: The correct answer is B.

In order for x to represent any extreme points in a polytope, there should be a vector such as $x \in S$ where $x = \lambda x_1 + (1 - \lambda)x_2$ with $x_1, x_2 \in S$ and $\lambda \in (0, 1)$.

Problem 11. Consider the following linear program:

$$\min \quad 3x_1 + 5x_2 + 2x_3 + 2x_4 \quad (1)$$

$$\text{s.t.} \quad x_1 - 3x_2 + 2x_3 + x_4 \leq 1 \quad (2)$$

$$x_1 + x_2 \leq 1 \quad (3)$$

$$-x_1 + x_2 \leq 1 \quad (4)$$

$$x_2 \geq 0 \quad (5)$$

$$0 \leq x_3 \leq 2 \quad (6)$$

$$0 \leq x_4 \leq 1 \quad (7)$$

Denote the polyhedron for variables x_1 and x_2 defined by constraints (3)-(5) as P_1 , and denote the polyhedron for x_3 and x_4 defined by constraints (6)-(7) as P_2 . Use two extreme points of $P_1 : (x_1^1, x_2^1) = (0, 1)$ and $(x_1^2, x_2^2) = (1, 0)$, and two extreme points of $P_2 : (x_3^1, x_4^1) = (0, 0)$ and $(x_3^2, x_4^2) = (2, 0)$. Which of the following is a correct restricted master problem of the linear program?

A

$$\min \quad 5\lambda_1 + 2\lambda_2 + 4\beta_2$$

$$\text{s.t.} \quad -3\lambda_1 + \lambda_2 + 4\beta_2 \leq 1$$

$$\lambda_1 + \lambda_2 \leq 1$$

$$\beta_1 + \beta_2 = 1$$

$$\lambda_1, \lambda_2 \geq 0$$

$$\beta_1, \beta_2 \geq 0.$$

B

$$\min \quad 5\lambda_1 + 2\lambda_2 + 2\beta_2$$

$$\text{s.t.} \quad -2\lambda_1 + \lambda_2 + 4\beta_2 \leq 1$$

$$\lambda_1 + \lambda_2 = 1$$

$$\beta_1 + \beta_2 = 1$$

$$\lambda_1, \lambda_2 \geq 0$$

$$\beta_1, \beta_2 \geq 0.$$

C

$$\begin{aligned}
& \min \quad 5\lambda_1 + 3\lambda_2 + 4\beta_2 \\
& \text{s.t.} \quad -3\lambda_1 + \lambda_2 + 4\beta_2 \leq 1 \\
& \quad \lambda_1 + \lambda_2 = 1 \\
& \quad \beta_1 + \beta_2 = 1 \\
& \quad \lambda_1, \lambda_2 \geq 0 \\
& \quad \beta_1, \beta_2 \geq 0.
\end{aligned}$$

Solution: C

First, note that the problem structure allows us to apply Dantzig-Wolfe decomposition. The key idea of Dantzig-Wolfe decomposition is that we can reformulate the problem using the fact that any point in a polytope can be represented as a convex combination of its extreme points. Then we can rule out option A, since the convexity constraint for P_1 , $\lambda_1 + \lambda_2$ should be equality, not inequality. Then, the objective value of the master problem should include the terms $\sum_{i=1}^2 \lambda_i (\mathbf{c}^\top \mathbf{x}^i) = \lambda_1 \begin{bmatrix} 3 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 5\lambda_1 + 3\lambda_2$. Thus the correct answer is C.

Problem 12. Which of the following points is the closest to $(4, 3, 2, 1)^\top \in \mathbb{R}^4$ in Euclidean norm?

- A $(3, 1, 2, 4)^\top$
- B $(4, 2, 1, 3)^\top$
- C $(3, 4, 1, 2)^\top$

Solution: C

Recall that the Euclidean distance (also known as the L_2 norm) between two points $\mathbf{x} = (4, 3, 2, 1)$ and $\mathbf{y}$ is given by the formula $\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}$. Then we can see that option C must be the closest point, since each of its components deviates from $\mathbf{x}$ by 1, and both of the other options have components that deviate from the components of $\mathbf{x}$ by ≥ 2 .

Problem 13. If we want to solve a least square problem $\min_x \|Ax - b\|_2^2$ where $A = \begin{pmatrix} 4 & 3 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$ and

$b = (4, 6, 2)^\top$, then

- A The optimal solution is $x = (2, -4/3)$.

- B The optimal solution is $x = (2, 4/3)$.
- C The optimal solution is $x = (-2, 4/3)$.
- D The optimal solution is $x = (-2, -4/3)$.

Solution: A

Minimizing the square of the L_2 norm is equivalent to minimizing the L_2 norm, since $f(x) = x^2$ is a monotonically increasing function for nonnegative x . From A and b , we can then see that we need to minimize $\sqrt{(4x_1 + 3x_2 - 4)^2 + (x_1 - 2)^2}$, since the other components of A are 0. We plug in the values from option A and get an objective of 0, which is the minimum possible value of the L_2 norm. Thus the solution is A.

Problem 14. If $A \in \mathbb{R}^{3 \times 3}$ is a symmetric matrix with singular values $\sigma_1 = 4$, $\sigma_2 = 1$, and $\sigma_3 = 0$, which of the following statements is true about the eigenvalues of A ?

- A The matrix A may not have eigenvalues.
- B The eigenvalues of A can be $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = 0$.
- C The eigenvalues of A can be $\lambda_1 = 4$, $\lambda_2 = 1$, and $\lambda_3 = 0$.

Solution: C

Recall from singular value decomposition (SVD) that if A is a symmetric matrix, SVD is almost equivalent to eigendecomposition. Namely, if A is symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$ and orthonormal eigenvectors $u_1, \dots, u_n$, i.e., $A = U\Lambda U^\top$, then $A = U\Sigma V^\top$, where $\sigma_i = |\lambda_i|$ and $v_i = \text{sign}(\lambda_i)$. Given the singular values in the problem, then, the eigenvalues of A are equal to the singular values of A , so the solution is C.

Problem 15. Consider the least square regression problem

$$z_1 = \min_{\beta_1, \dots, \beta_n} \sum_{i=1}^N (y_i - \sum_{j=1}^n \beta_j x_{ij})^2$$

and the absolute deviation regression problem defined as

$$z_2 = \min_{\beta_1, \dots, \beta_n} \sum_{i=1}^N |y_i - \sum_{j=1}^n \beta_j x_{ij}|$$

where $x_{ij} (i = 1, \dots, N; j = 1, \dots, n); y_i (i = 1, \dots, N)$ are training data. Which of the following statements is true?

- A z_1 and z_2 are always equal.

B We cannot solve the absolute deviation regression problem by linear programming.

C The optimal solutions can be different.

Solution: C

Clearly, option A is false, since the formula for z_1 is that of z_2 squared; even if the values of β_i coincide, the objective values z_1, z_2 will be different. Then option C must be true, since options A and C are mutually exclusive. For posterity, we evaluate option B and we determine that option B is false, since we can reformulate problems involving absolute values into linear programs. Thus the solution is C.

Problem 16. Is $\begin{pmatrix} 3 \\ -3 \\ 5 \end{pmatrix} \succeq_{\mathbb{R}_+^3} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$?

A Yes

B No

Solution: B

Since

$$\begin{pmatrix} 3 \\ -3 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \notin \mathbb{R}_+^3$$

correct answer is "No" i.e. B.

Problem 17. Which of the following cannot be formulated as a second-order cone constraint?

A $x^2 + y^2 - 2x + 4y \leq 1$

B $x^2 - y^2 - 2x + 4y \leq 1$

C $x^2 + y^2 \geq 1$

D $xy - z^2 \geq 0$

Solution: C

We can reformulate constraints A, B, and D into second-order cone constraints:

$$x^2 + y^2 - 2x + 4y \leq 1 \quad \Leftrightarrow \quad (x-1)^2 + (y-2)^2 \leq 6$$

$$x^2 - y^2 - 2x + 4y \leq 1 \quad \Leftrightarrow \quad (x-1)^2 + 2 \leq (y-2)^2$$

$$xy - z^2 \geq 0 \quad \Leftrightarrow \quad \left(\frac{x-y}{2}\right)^2 + z^2 \leq \left(\frac{x+y}{2}\right)^2$$

Thus the only constraint which can't be reformulated as second-order cone constraint is C.

Problem 18. Which of the following matrices is symmetric positive semidefinite?

A $\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$

B $\begin{bmatrix} 3 & -2 \\ 2 & 4 \end{bmatrix}$

C $\begin{bmatrix} -6 & -5 \\ -5 & -3 \end{bmatrix}$

D $\begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix}$

Solution: D

Matrix B is not symmetric, so that is not the correct answers. Matrices A, C, and D are symmetric. Let's see which one of those three is positive semidefinite.

Option 1: Let's use Sylvester's criterion. First leading principal minor of matrix C is -6, thus it is not positive semidefinite. First leading principal minor of matrix A is $2 \geq 0$, but its second leading principal minor is

$$\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -2 - 1 = -3 < 0$$

Thus matrix A is not positive semidefinite. Finally, first principal minor of matrix D is $3 \geq 0$, and its second leading principal minor is

$$\begin{vmatrix} 3 & -2 \\ -2 & 4 \end{vmatrix} = 12 + 4 = 16 \geq 0.$$

Thus matrix D is positive semidefinite, further more it is positive definite.

Option 2: We can find eigenvalues for each matrix, and show that only matrix D has nonnegative eigenvalues

Problem 19. Given a hyperplane $H : 6x_1 - 8x_2 = -1$ and a point $p = [-1, 1]$, find the distance between H and p .

A $\frac{3}{2}$

B $\frac{2}{3}$

C $\frac{13}{10}$

D $\frac{10}{13}$

Solution: C

Using formula:

$$d(a^\top x - b, p) = \frac{|a^\top p - b|}{\|a\|_2}$$

we get

$$d(H, p) = \frac{|6(-1) - 8(1) + 1|}{\sqrt{6^2 + 8^2}} = \frac{13}{10}$$

so correct answer is C.

Problem 20. How many possible values can an integer variable x take where $x \in [0.5, 3.5]^2 \cap \mathbb{Z}^2$?

A 4

B 9

C 10

D 16

Solution: B

Since $x = (x_1, x_2) \in [0.5, 3.5]^2 \cap \mathbb{Z}^2 = \{1, 2, 3\}^2$, we have 3 possible values for x_1 and 3 possible values for x_2 . Thus total number of possible values for x is $3 \cdot 3 = 9$, i.e. correct answer is B.

Problem 21. “If items 1 and 3 are chosen, then item 2 is chosen” can be modeled using binary variables $x_1, x_2, x_3 \in \{0, 1\}$ as:

A $x_2 \leq x_1, x_2 \leq x_3$

B $x_2 \geq x_1 + x_3 - 1$

C $x_1 \geq \frac{(x_2 + x_3)}{2}$

D $x_2 \leq x_1 \leq x_3$

Solution: B

Option A is true for $x_1 = 1, x_2 = 0, x_3 = 1$, but the desired constraint should be false.

Option B is correct.

Option C is false for $x_1 = 0, x_2 = 1, x_3 = 1$, but the desired constraint should be true.

Option D is true for $x_1 = 1, x_2 = 0, x_3 = 1$, but the desired constraint should be false.

Problem 22. “If item 2 is chosen, then item 1 or 3 is chosen” can be modeled using binary variables $x_1, x_2, x_3 \in \{0, 1\}$ as:

A $x_2 \leq x_1, x_2 \leq x_3$

B $x_2 \leq x_1 + x_3$

C $x_2 \leq x_1, 1 - x_2 \leq x_3$

D $x_2 \leq x_1, 1 - x_1 \leq x_3$

Solution: B

Option A is false for $x_1 = 1, x_2 = 1, x_3 = 0$, but the desired constraint should be true.

Option B is correct.

Option C is false for $x_1 = 0, x_2 = 1, x_3 = 1$, but the desired constraint should be true.

Option D is false for $x_1 = 0, x_2 = 1, x_3 = 1$, but the desired constraint should be true.

Problem 23. “If item 2 is not chosen, then item 1 and item 3 are chosen” can be modeled using binary variables $x_1, x_2, x_3 \in \{0, 1\}$ as:

A $x_2 \leq x_1 \leq x_3$

B $x_1 + x_2 \leq 1, x_2 + x_3 \leq 1$

C $x_1 + x_2 \geq 1, x_2 + x_3 \geq 1$

D $x_2 \leq x_1, x_2 \leq x_3$

Solution: C

Option A is false for $x_1 = 0, x_2 = 1, x_3 = 0$, but the desired constraint should be true.

Option B is true for $x_1 = 1, x_2 = 0, x_3 = 0$, but the desired constraint should be false.

Option C is correct.

Option D is true for $x_1 = 1, x_2 = 0, x_3 = 0$, but the desired constraint should be false.

Problem 24. “If item 2 or item 3 is chosen, then item 1 is chosen” can be modeled using binary variables $x_1, x_2, x_3 \in \{0, 1\}$ as:

A $x_1 \leq x_2 + x_3$

B $x_1 \leq x_2 \leq x_3$

C $x_1 \geq x_2, x_3 \geq x_1$

D $2x_1 \geq x_2 + x_3$

Solution: D

Option A is true for $x_1 = 0, x_2 = 0, x_3 = 1$, but the desired constraint should be false.

Option B is true for $x_1 = 0, x_2 = 0, x_3 = 1$, but the desired constraint should be false.

Option C is true for $x_1 = 0, x_2 = 0, x_3 = 1$, but the desired constraint should be false.

Option D is correct.

Problem 25. “Either $x_1 + 3x_2 \leq 3$ or $4x_1 - x_2 \geq 1$ ” can be formulated as $y \in \{0, 1\}$, $x_1 + 3x_2 \leq 3 + My$ and

A $4x_1 + x_2 \geq 1 + My$

B $x_1 - x_2 \leq 2 - M(1 - y)$

C $4x_1 - x_2 \geq 1 - M(1 - y)$

D $4x_1 - x_2 \geq M(1 - y)$

Solution: C

If $y = 0$ then the first constraint is active. That means the second constraint must be inactive. A does not make sense. That does not relax the constraint. B also does not make much sense. The left side of the inequality does not match. C does make sense as that would relax the constraint. D definitely does not relax the constraint. As such, C is the only remaining option. One could try to see what would happen if $y = 1$. That would require the second constraint to be active and again, we see that C does this.

Problem 26. “If $5x_1 + x_2 > 5$, then $x_1 - 2x_2 + x_3 \geq 8$ ” can be formulated as $y \in \{0, 1\}$, $x_1 - 2x_2 + x_3 \geq 8 - M(1 - y)$ and

A $5x_1 + x_2 \geq 5 + M(1 - y)$

B $5x_1 + x_2 \leq 5 + My$

C $5x_1 + x_2 > 5 - M(1 - y)$

Solution: B

We can see that A right off the bat does not work. When $y = 0$, the constraint becomes $5x_1 + x_2 \geq 5 + M$ which does not make sense.

C looks promising, however the issue is that even when $5x_1 + x_2 > 5$ is true, y can still be 0 as well as 1. And as such, the second constraint may not be active.

For B, when $y = 0$, then $5x_1 + x_2 > 5$ is not true and as such $x_1 - 2x_2 + x_3 \geq 8$ is not active. When $y = 1$ which is when $5x_1 + x_2 > 5$ is true, then only does $x_1 - 2x_2 + x_3 \geq 8$ become active.

Problem 27. Consider the assignment problem

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^m x_{ij} = 1, \quad j = 1, \dots, n, \\ & \sum_{j=1}^n x_{ij} \leq 1, \quad i = 1, \dots, m, \\ & x_{ij} \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n \end{aligned}$$

where i is the index for workers and j is the index for jobs. If we are going to model this problem as a network flow problem, how many nodes are supply nodes and how many nodes are demand nodes?

- A $m + 1$ supply nodes and n demand nodes
- B n supply nodes and m demand nodes
- C n supply nodes and $m + 1$ demand nodes
- D m supply nodes and $n + 1$ demand nodes

Solution: D

There are m supply nodes and $n + 1$ demand nodes. The m supply nodes is fairly self explanatory and the extra demand node is the slack node. This is mentioned in Module 27, Lesson 12, Slide 13.

Problem 28. The number of possible tours in a TSP with 4 cities is at most

- A 24
- B 1024
- C 6

Solution: A

You start off at any one of four cities. At that point you could travel to any one of the three remaining cities, then the two remaining cities until you visit the last one. That becomes $4!$ which is 24.

Problem 29. We have a set of surgeons and a set of surgery types. Each surgeon can do a subset of the surgery types. We need to pick a subset of surgeons such that, for each surgery type, there is at least one surgeon in the selected set that can do it. This is an example of a

- A Set packing problem
- B Set covering problem
- C Set partitioning problem

Solution: B

Because at least one surgeon can be selected, that eliminates the Set partitioning problem in which it needs to be exactly one surgeon for each surgery type. And we are looking to make sure that at least one surgeon in the set can cover the surgery type, not the other way around. Set packing also does not make sense. This leaves us with Set Covering.

Problem 30. Which of the following statements is true?

- A A minimum cost network flow problem with integer data with an optimal solution has an integer optimal solution
- B A minimum cost network flow problem always has an optimal solution
- C A minimum cost network flow problem with arbitrary data always has an integer optimal solution

Solution: A

A is shown to be true in Module 27, Lesson 2, Slide 9. If the supply data and the capacities are integers, then every extreme point will also be an integer.

At no point is it stated that a minimum cost network flow problem has to have an optimal solution, nor anything about arbitrary data.